

Henil Parmar

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EDUCATION

Indian Institute of Technology Bombay
Bachelor of Technology in Aerospace Engineering

[Jul '23– May '27]
CPI: 7.42 / 10 (Ongoing)

Key Courses: Control Theory, Flight Mechanics of Aircrafts and Spacecrafts, Introduction to Navigation and Guidance, State Space Methods for Flight Vehicles, Sensors and Actuators, Aircraft Design

PROFESSIONAL EXPERIENCE

Aerospace Engineering Intern | *Airbus Defence and Space*
C295 Make in India Program, Final Assembly Line, Vadodara

[Jun '26–Aug '26]

- Analyzed **C295** Final Assembly Line workflows, mastering station-wise ops. & complex system integration context
- Validated assembly quality across multiple production stages utilizing advanced **aerospace test equipment** onsite
- Identified process bottlenecks and learned **lean manufacturing solutions** to optimize overall FAL productivity
- Streamlined large-scale assembly ops. via **cross-functional coordination** across Engineering and Quality teams
- Secured **FAL Program Manager LOR** for outstanding performance across C295 final assembly line workflows

Summer Research Intern | *Space Applications Centre, ISRO*

Reusable Launch Vehicle guidance-algorithm development under Dr. Ashish Kumar Shukla

[May '25–Jul '25]

- Developed AI-powered guidance algorithm via **GPS, INS** for precise, high-accuracy **RLV autonomous landings**
- Optimized **Pseudolite navigation** using **Monte Carlo** simulations to enhance robustness of guidance for RLV
- Formulated **multi-modal guidance algorithm** establishing precise, true autonomous landing capabilities for RLV
- Integrated mathematical models into visual simulations, presenting control advancements to senior scientists well
- Earned **LORs from Scientist & SAC former Director** for novel algorithm design and simulation achievements

RESEARCH EXPERIENCE

Reusable Launch Vehicle (RLV) Design | *Supervised Learning Project*
Aerospace Vehicle Design Series (~1.5 years) | Guide: Prof. Dhwani Shukla

[Sep '24–Nov '25]

- Investigated **RLV return & landing-gear** in team of 4 using mission trajectory and design under faculty guidance
- Designed **G-FOLD & terminal-velocity** guidance for sub-2 m/s controlled descent of 549t LEO Launch Vehicle
- Evaluated **7.2%** return fuel allocation for **2,858 m/s Δv target** linking guidance to landing-system constraints
- Engineered 4 landing legs in AutoCAD for **472.4 kN** impact, linking structural & guidance constraints effectively

KEY PROJECTS

F1 Minimum-Lap-Time Simulator | *Self Project*

Vehicle dynamics, optimal control and telemetry-based validation

[Nov '25–Dec '25]

- Developed Python min-lap-time optimal-control simulator in **CasADi/IPOPT** using degree-three **Radau direct collocation**, jointly optimizing racing line and speed profile across 6 real F1 circuit centerlines for realistic analysis
- Implemented open-wheel vehicle physics with quadratic drag, power limits, load transfer, and tyre-force constraints
- Integrated **FastF1 telemetry** evaluating lap-time error, speed RMSE, cumulative time delta, sectors/corners & braking zones; benchmarked 2021 Monza at **+6.28% lap-time error** and **24.29 km/h speed RMSE** uncalibrated

Satellite Attitude Control using LQR | *Course Project*

State Space Methods for Flight Vehicles | Guide: Prof. Shashi Ranjan Kumar

[Aug'26– Present]

- Constructed 3-axis satellite attitude dynamics as 6-state model, linearized around nominal equilibrium for control
- Assessed **controllability & observability**, designing an **LQR state feedback law** for satellite attitude regulation
- Simulated closed-loop satellite response, tuned **LQR weights** to improve settling, tracking accuracy & control effort

Vehicle State Estimation & Nonlinear MPC | *Self Project*

Nonlinear vehicle dynamics, sensor fusion and constrained feedback control

[Sep '26–Present]

- Simulated **6-state nonlinear bicycle dynamics** with tire saturation, aero loading, and control constraints for F1
- Executed **multi-sensor EKF fusion** using GPS, IMU, & wheel speeds; preserved state estimation via 4 s GPS loss
- Deployed CasADi/IPOPT **NMPC** via EKF feedback under track/steering limits; hit 1.49m lateral RMS, 0 violations

Hybrid Energy Management Optimizer | *Self Project* [May '26–Jun '26]
Optimal control, hybrid powertrain modelling and 2026 FIA energy-management optimization

- Framed optimal control in CasADi, leveraging **IPOPT** interior-point solver for MGU-K strategy across 4 F1 circuits
- Encoded **2026 FIA** hybrid constraints: **350 kW MGU-K**, **7 MJ** recharge with aero, tyre & ICE-electric coupling
- Benchmarked 3 energy strategies via multi-start mesh convergence, FastF1 telemetry & **42-test verification suite**

Sensor Fusion for Human Motion State Estimation | *Course Project* [Jan'26–Apr'26]
Joint Biomechanics | Guide: Prof. Darshan Shah

- Built **MediaPipe Pose** 2D vision pipeline to extract hip, knee, ankle keypoints from 24 images over 6 test subjects
- Correlated pose with accelerometer data across 4 static poses in 6 subjects, achieving angle accuracy of $\mathbf{r} = \mathbf{0.984}$
- Validated markerless pose estimates against goniometers, achieving $\mathbf{r} = \mathbf{0.990}$ agreement for **computed moments**

STOL Aircraft & Airfoil Design | *Course Project* [Jan'24– Apr'24]
Low Speed Aerodynamics | Guide: Prof. Dhwanil Shukla

- Generated **ANSYS Fluent** 2D subsonic CFD meshes for **NACA 5311**, attaining peak C_l/C_d of **61.11** at 15° stall
- Co-developed Python tool to compute C_l , circulation & induced velocity via Thin Airfoil Theory for custom airfoils
- Modeled STOL aerodynamics in **OpenVSP** using empirical drag methods for **157 m** takeoff, boosting C_l by **8.3%**

Electrostatic Pull-In Analysis of MEMS Parallel-Plate Actuator | *Course Project* [Aug'26– Present]
Sensors and Actuators | Guide: Prof. Pradeep Dixit

- Calculated analytical **MEMS pull-in voltage** via electrostatic and spring force equilibrium at instability threshold
- Executed coupled **electrostatic-structural FEA** for plate deflection, nonlinear instability & onset pull-in voltage
- Validated analytical pull-in predictions against FEA using key gap, plate-area and spring-stiffness parametric sweeps

Universal Testing Machine | *Course Project* [Aug'23–Nov'23]
Makerspace | Guides: Prof. Joseph John, Prof. K.N. Jonnalagadda

- Co-engineered Universal Testing Machine in 6-person team to measure test specimen **Young's modulus** empirically
- Designed & built **lead-screw transmission system** to apply controlled mechanical loads across multiple samples
- Automated test data generation by integrating force-displacement tracking via **Arduino UNO**, **IR sensor**, **PWM**

TECHNICAL SKILLS

Languages & Shell	Python, C++, MATLAB, PowerShell, MySQL, HTML, L ^A T _E X
Optimization & Simulation	CasADi, IPOPT, CVXPY, ECOS, ANSYS (Fluent, Structural), Simulink, OpenVSP
Scientific & Data Libraries	NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-Learn, PyTorch, LapSim, pdfplumber, openpyxl, PyYAML
CAD & Developer Tools	SolidWorks, Fusion 360, LaserCAD, Arduino IDE, MS Excel, Streamlit, Subprocess/System Scripting

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Joint Secretary | *Aerospace Engineering Department, IIT Bombay* [Apr '25–Apr '26]

- Led 8-member council managing events, workshops, academic coordination & faculty engagement for **450+** students
- Contributed to development of the department website and digital outreach activities to boost student engagement

Student Mentor | *Department Academic Mentor Program, IIT Bombay* [Jul '25–Present]

- Selected as one of **28 mentors** competitively; provided academic & social guidance to 2nd-year aerospace students
- Awarded **Best DAMP Mentor** among 28 peers for academic guidance & exceptional student support in 2026
- Appointed as **Sub-Group Head** of Policies for DAMP; leading curriculum support for Aerospace Engineering

System Administrator | *Hostel 16 Council, IIT Bombay* [Dec '23–May '24]

- Coordinated with IITB Computer Centre to deploy IITB Wireless and optimize LAN network infrastructure for **1,000+ residents**, delivering direct network and technical computing support across hostel residential blocks

Appointed as mentor for the **Academic Support Program (ASP/ARP)**; provided structured academic guidance, study strategy interventions, and targeted peer support to assist students in recovering and clearing core backlogs

Additional service: Served as Academic Coach at Vidhyadaan, supporting **20+ youth** through literacy and learning initiatives; volunteered with NSS to deliver academic support to underprivileged students